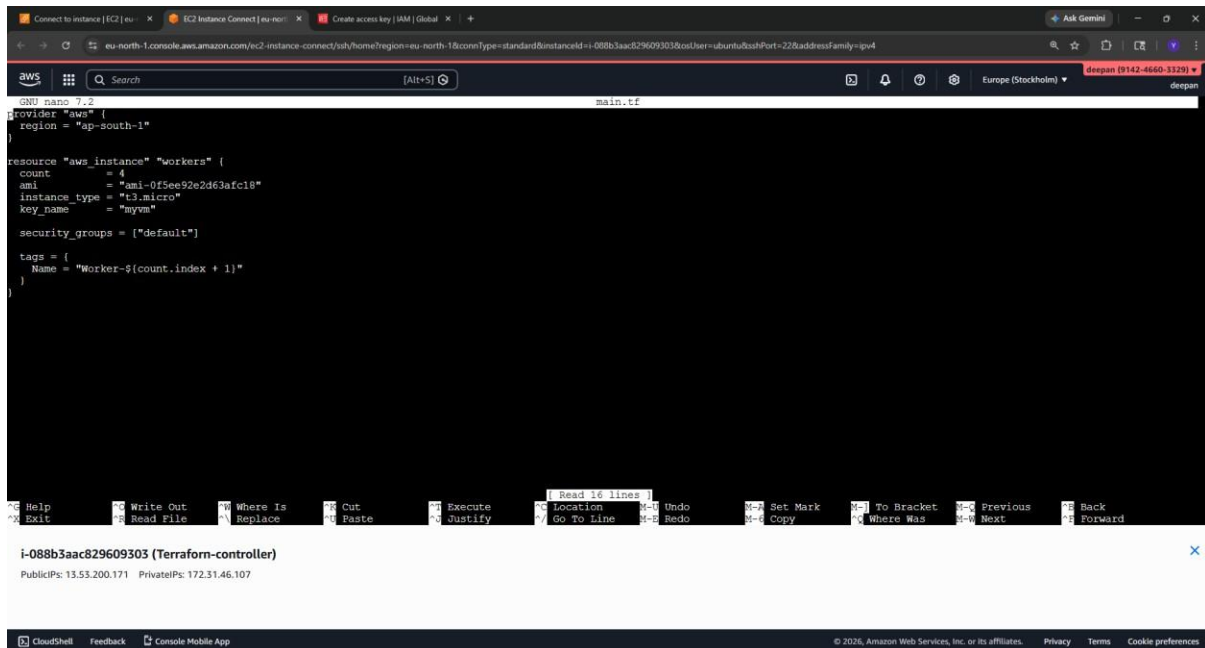


SCREENSHOTS

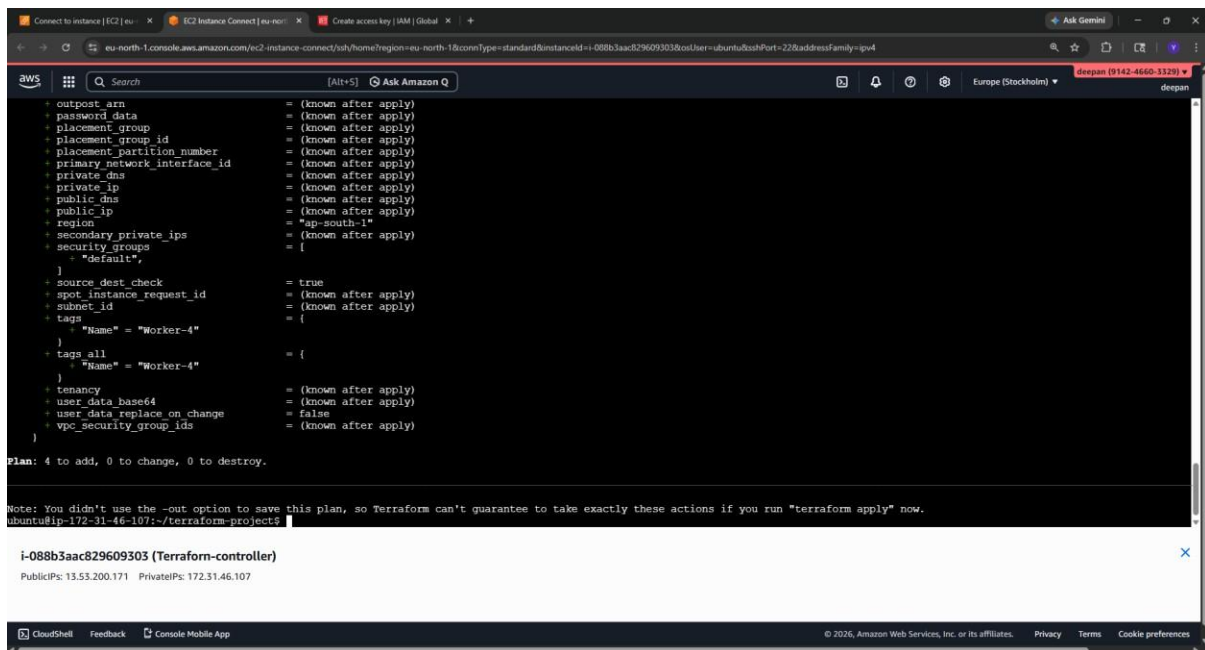


The screenshot shows the AWS CloudShell interface. The terminal displays a Terraform configuration file named `main.tf`. The configuration defines an AWS provider for the `eu-south-1` region and a resource `aws_instance.workers` with the following details:

- `count`: 4
- `ami`: `"ami-0f5ee92e2d63afc18"`
- `instance_type`: `"t3.micro"`
- `key_name`: `"mym"`
- `security_groups`: `["default"]`
- `tags`: `{ Name = "Worker-${count.index + 1}" }`

Below the configuration, the instance ID `i-088b3aac829609303` is shown, identified as the `Terraform-controller`. It lists public IP addresses `13.53.200.171` and private IP addresses `172.31.46.107`. The bottom of the interface shows the AWS CloudShell footer with copyright information for 2026.

Using Terraform, accomplish the task of infrastructure creation in the AWS cloud provider



The screenshot shows the AWS CloudShell interface displaying the output of a `terraform plan` command. The plan lists the following resources to be added:

- `outpost_arn`: `= (known after apply)`
- `password_data`: `= (known after apply)`
- `placement_group`: `= (known after apply)`
- `placement_group_id`: `= (known after apply)`
- `placement_partition_number`: `= (known after apply)`
- `primary_network_interface_id`: `= (known after apply)`
- `private_dns`: `= (known after apply)`
- `private_ip`: `= (known after apply)`
- `public_dns`: `= (known after apply)`
- `public_ip`: `= (known after apply)`
- `region`: `= "ap-south-1"`
- `secondary_private_ips`: `= (known after apply)`
- `security_groups`: `= [ "default", "default" ]`
- `source_dest_check`: `= true`
- `spot_instance_request_id`: `= (known after apply)`
- `subnet_id`: `= (known after apply)`
- `tags`: `= { "Name" = "Worker-4" }`
- `tags_all`: `= { "Name" = "Worker-4" }`
- `tenancy`: `= (known after apply)`
- `user_data_base64`: `= (known after apply)`
- `user_data_replace_on_change`: `= false`
- `vpc_security_group_ids`: `= (known after apply)`

The plan summary indicates: `Plan: 4 to add, 0 to change, 0 to destroy.`

A note at the bottom states: `Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.`

The instance ID `i-088b3aac829609303` is shown, identified as the `Terraform-controller`. It lists public IP addresses `13.53.200.171` and private IP addresses `172.31.46.107`. The bottom of the interface shows the AWS CloudShell footer with copyright information for 2026.

```
aws
+ subnet_id = (known after apply)
+ tags = {
+   "Name" = "Worker-4"
+ }
+ tags_all = {
+   "Name" = "Worker-4"
+ }
+ tenancy = (known after apply)
+ user_data_base64 = (known after apply)
+ user_data_replace_on_change = false
+ vpc_security_group_ids = (known after apply)
}

Plan: 4 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_instance.workers[1]: Creating...
aws_instance.workers[3]: Creating...
aws_instance.workers[0]: Creating...
aws_instance.workers[2]: Creating...
aws_instance.workers[3]: Still creating... [10s elapsed]
aws_instance.workers[1]: Still creating... [10s elapsed]
aws_instance.workers[0]: Still creating... [10s elapsed]
aws_instance.workers[2]: Still creating... [10s elapsed]
aws_instance.workers[0]: Creation complete after 12s [id=i-046491f38a1d58b07]
aws_instance.workers[2]: Creation complete after 12s [id=i-0b2f99bc9a9d3d7f]
aws_instance.workers[1]: Creation complete after 12s [id=i-0b175896d7f855fcd]
aws_instance.workers[3]: Creation complete after 12s [id=i-0e9a1eec24b2c2714]

Apply complete! Resources: 4 added, 0 changed, 0 destroyed.
ubuntu@ip-172-31-46-107:~/terraform-project$
```

i-088b3aac829609303 (Terraform-controller)

PublicIPs: 13.53.200.171 PrivateIPs: 172.31.46.107

Instances | EC2 | eu-north-1

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#instances

Search [Alt+S] Ask Amazon Q

Europe (Stockholm) deepan (9142-4660-3329) deepan

EC2 > Instances

EC2

Dashboard

AWS Global View

Events

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Successfully attached Allaccess to instance i-088b3aac829609303

Instances (1/5) info

Save filter sets Choose filter set Find Instance by attribute or tag (case-sensitive)

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ..
☒	Terraform-cont...	i-088b3aac829609303	Running	t3.micro	3/3 checks pass	View alarms +	eu-north-1b	ec2-13-53-200-171.eu-...	13.53.200.17
☐	Worker-4	i-0e9a1eec24b2c2714	Running	t3.micro	Initializing	View alarms +	eu-north-1b	ec2-13-49-68-7.eu-nort...	13.49.68.7
☐	Worker-1	i-046491f38a1d58b07	Running	t3.micro	Initializing	View alarms +	eu-north-1b	ec2-13-63-237-146.eu-...	13.63.237.146
☐	Worker-2	i-0b175896d7f855fcd	Running	t3.micro	Initializing	View alarms +	eu-north-1b	ec2-13-60-196-107.eu-...	13.60.196.107
☐	Worker-3	i-0b2f99bc9a9d3d7f	Running	t3.micro	Initializing	View alarms +	eu-north-1b	ec2-16-171-31-247.eu-...	16.171.31.247

i-088b3aac829609303 (Terraform-controller)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary info

Instance ID

i-088b3aac829609303

Public IPv4 address

13.53.200.171 | open address

Private IPv4 addresses

172.31.46.107

Instance state

Running

Public DNS

ec2-13-53-200-171.eu-north-1.compute.amazonaws.com | open address

Architectural Advice

Instances [EC2 [eu-north-1] x EC2 Instance Connect [eu-north-1] x EC2 Instance Connect [eu-north-1] x EC2 Instance Connect [eu-north-1] x EC2 Instance Connect [eu-north-1] x EC2 Instance Connect [eu-north-1] x +

eu-north-1.console.aws.amazon.com/ec2-instance-connect/home?region=eu-north-1&connType=standard&instanceId=i-0b2f99bc9a9d32d7f&osUser=ubuntu&sshPort=22&addressFamily=ipv4

Search [Alt+S]

Europe (Stockholm) deepan

No VM guests are running outdated hypervisor (qemu) binaries on this host.
kubelet set on hold.
kubeadm set on hold.
kubectl set on hold.
ubuntu@ip-172-31-38-74:~\$ sudo swapoff -a
ubuntu@ip-172-31-38-74:~\$ sudo modprobe br_netfilter

echo '1' | sudo tee /proc/sys/net/bridge/bridge-nf-call-iptables
1
ubuntu@ip-172-31-38-74:~\$ sudo systemctl start docker
ubuntu@ip-172-31-38-74:~\$ sudo kubeadm join 172.31.44.185:6443 --token seq8gs.a6p83z5mr647vxgy \\\n--discovery-token-ca-cert-hash sha256:4645b72a07c7836bb43866a8049183b1662349d8f9f58412c88441d8cefa53c8
[preflight] Running pre-flight checks
[preflight] Reading configuration from the cluster...
[preflight] FYI: You can look at this config file with 'kubectl -n kube-system get cm kubeadm-config -o yaml'
[kubeadm-start] Writing kubelet configuration to file "/var/lib/kubelet/config.yaml"
[kubeadm-start] Writing kubelet environment file with flags to file "/var/lib/kubelet/kubeadm-flags.env"
[kubeadm-start] Starting the kubelet
[kubeadm-check] Waiting for a healthy kubelet at http://127.0.0.1:10248/healthz. This can take up to 4m0s
[kubeadm-check] The kubelet is healthy after 1.00600764s
[kubeadm-start] Waiting for the kubelet to perform the TLS Bootstrap

This node has joined the cluster:
* Certificate signing request was sent to apiservert and a response was received.
* The Kubelet was informed of the new secure connection details.

Run 'kubectl get nodes' on the control-plane to see this node join the cluster.

ubuntu@ip-172-31-38-74:~\$

i-0b2f99bc9a9d32d7f (Worker-3)

PublicIPs: 16.171.31.247 PrivateIPs: 172.31.38.74

CloudShell Feedback Console Mobile App

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Instances [EC2 [eu-north-1] x EC2 Instance Connect [eu-north-1] x EC2 Instance Connect [eu-north-1] x EC2 Instance Connect [eu-north-1] x EC2 Instance Connect [eu-north-1] x EC2 Instance Connect [eu-north-1] x +

eu-north-1.console.aws.amazon.com/ec2-instance-connect/home?region=eu-north-1&connType=standard&instanceId=i-0e9a1eec24b2c2714&osUser=ubuntu&sshPort=22&addressFamily=ipv4

Search [Alt+S]

Europe (Stockholm) deepan

No VM guests are running outdated hypervisor (qemu) binaries on this host.
kubelet set on hold.
kubeadm set on hold.
kubectl set on hold.
ubuntu@ip-172-31-43-13:~\$ sudo swapoff -a
ubuntu@ip-172-31-43-13:~\$ sudo modprobe br_netfilter

echo '1' | sudo tee /proc/sys/net/bridge/bridge-nf-call-iptables
1
ubuntu@ip-172-31-43-13:~\$ sudo systemctl start docker
ubuntu@ip-172-31-43-13:~\$ sudo kubeadm join 172.31.44.185:6443 --token seq8gs.a6p83z5mr647vxgy \\\n--discovery-token-ca-cert-hash sha256:4645b72a07c7836bb43866a8049183b1662349d8f9f58412c88441d8cefa53c8
[preflight] Running pre-flight checks
[preflight] Reading configuration from the cluster...
[preflight] FYI: You can look at this config file with 'kubectl -n kube-system get cm kubeadm-config -o yaml'
[kubeadm-start] Writing kubelet configuration to file "/var/lib/kubelet/config.yaml"
[kubeadm-start] Writing kubelet environment file with flags to file "/var/lib/kubelet/kubeadm-flags.env"
[kubeadm-start] Starting the kubelet
[kubeadm-check] Waiting for a healthy kubelet at http://127.0.0.1:10248/healthz. This can take up to 4m0s
[kubeadm-check] The kubelet is healthy after 501.585455ms
[kubeadm-start] Waiting for the kubelet to perform the TLS Bootstrap

This node has joined the cluster:
* Certificate signing request was sent to apiservert and a response was received.
* The Kubelet was informed of the new secure connection details.

Run 'kubectl get nodes' on the control-plane to see this node join the cluster.

ubuntu@ip-172-31-43-13:~\$

i-0e9a1eec24b2c2714 (Worker-4)

PublicIPs: 13.49.68.7 PrivateIPs: 172.31.43.13

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The screenshot shows the AWS CloudShell terminal interface. The terminal output displays the process of pulling Docker images for 'nginx:latest' and building a Docker image. The build process is successful, resulting in the image '151e55a644a8'. Below the terminal output, the instance details are shown: 'i-Ob175896d7f855fcd (Worker-2)' with PublicIPs: 13.60.52.77 and PrivateIPs: 172.31.44.185. The bottom of the screenshot shows the AWS CloudShell navigation bar with 'CloudShell', 'Feedback', and 'Console Mobile App' links, along with the copyright notice '© 2016, Amazon Web Services, Inc. or its affiliates.' and links for 'Privacy', 'Terms', and 'Cookie preferences'.

```
84e114c2bb36: Pulling fs layer
8435b2dcdf5c: Pulling fs layer
054715a6bffa: Pulling fs layer
4a038fd18db1: Pulling fs layer
88d1d984b765: Pulling fs layer
7b5d674621c2: Pulling fs layer
7b5d674621c2: Download complete
84e114c2bb36: Download complete
448ea5cac5d5: Download complete
4a038fd18db1: Download complete
88d1d984b765: Download complete
3eff0a97d435: Download complete
cc0cf959117b: Download complete
5435b2dcdf5c: Download complete
054715a6bffa: Download complete
5435b2dcdf5c: Pull complete
7b5d674621c2: Pull complete
84e114c2bb36: Pull complete
4a038fd18db1: Pull complete
88d1d984b765: Pull complete
054715a6bffa: Pull complete
448ea5cac5d5: Pull complete
Digest: sha256:7f0adcalfc6c29c8dc49a2e90037a10ba20dc266baaed0988e9fb4d0d8b85ba0
Status: Downloaded newer image for nginx:latest
--> 7f0adcalfc6c
Step 2/2 : COPY . /usr/share/nginx/html
--> 151e55a644a8
Successfully built 151e55a644a8
Successfully tagged yourdockerhubusername/website:v1
ubuntu@ip-172-31-44-185:~/website$
```

i-Ob175896d7f855fcd (Worker-2)
PublicIPs: 13.60.52.77 PrivateIPs: 172.31.44.185

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Installing respective services

The screenshot shows the AWS CloudShell terminal interface. The terminal output displays the process of pushing the Docker image 'despan2003/website:v1' to the Docker Hub repository. The push is successful. Below the terminal output, the instance details are shown: 'i-Ob175896d7f855fcd (Worker-2)' with PublicIPs: 13.60.52.77 and PrivateIPs: 172.31.44.185. The bottom of the screenshot shows the AWS CloudShell navigation bar with 'CloudShell', 'Feedback', and 'Console Mobile App' links, along with the copyright notice '© 2016, Amazon Web Services, Inc. or its affiliates.' and links for 'Privacy', 'Terms', and 'Cookie preferences'.

```
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
The push refers to repository [docker.io/yourdockerhubusername/website]
84e114c2bb36: Waiting
7b5d674621c2: Waiting
448ea5cac5d5: Waiting
054715a6bffa: Waiting
88d1d984b765: Waiting
push access denied, repository does not exist or may require authorization: server message: insufficient_scope: authorization failed
The push refers to repository [docker.io/deспан2003/website]
tag does not exist: desпан2003/website:v1
ubuntu@ip-172-31-44-185:~/website$ docker build -t desпан2003/website:v1 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit.
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  257kB
Step 1/2 : FROM nginx:latest
--> 7f0adcalfc6c
Step 2/2 : COPY . /usr/share/nginx/html
--> Using cache
--> 151e55a644a8
Successfully built 151e55a644a8
Successfully tagged desпан2003/website:v1
ubuntu@ip-172-31-44-185:~/website$ docker push desпан2003/website:v1
The push refers to repository [docker.io/deспан2003/website]
88d1d984b765: Mounted from library/nginx
5435b2dcdf5c: Mounted from library/nginx
054715a6bffa: Mounted from library/nginx
84e114c2bb36: Mounted from library/nginx
4a038fd18db1: Mounted from library/nginx
7b5d674621c2: Mounted from library/nginx
448ea5cac5d5: Mounted from library/nginx
88d1d984b765: Pushed
v1: digest: sha256:151e55a644a8b880bbdb3500839a96db1c7f5b9c2a9fc586a5fb81f8a456e8 size: 2046
ubuntu@ip-172-31-44-185:~/website$
```

i-Ob175896d7f855fcd (Worker-2)
PublicIPs: 13.60.52.77 PrivateIPs: 172.31.44.185

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The code should be containerized with the help of the Dockerfile. The Dockerfile should be built every time if there is a push to GitHub. Create a custom Docker image using a Dockerfile.

```
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 257kB
Step 1/2 : FROM nginx:latest
--> 7f0adcafc6c
Step 2/2 : COPY . /usr/share/nginx/html
--> Using cache
--> 151e55a644a8
Successfully built 151e55a644a8
Successfully tagged deepan2003/website:v1
ubuntu@ip-172-31-44-185:~/website$ docker push deepan2003/website:v1
The push refers to repository [docker.io/deepan2003/website]
88d1d984b765: Mounted from library/nginx
5435b2dcdf5c: Mounted from library/nginx
054715a6dffa: Mounted from library/nginx
84e114c2bb36: Mounted from library/nginx
4a038fd18db1: Mounted from library/nginx
7b5d674621c2: Mounted from library/nginx
448ea5cac5d5: Mounted from library/nginx
98e94c91c91c: Pushed
v1: digest: sha256:151e55a644a8b6a880b3b500839a96db1c7f5b9c2a9fc586a5fb81f8a456e8 size: 2046
ubuntu@ip-172-31-44-185:~/website$ nano deployment.yaml
deployment.apps/website-deployment created
ubuntu@ip-172-31-44-185:~/website$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
website-deployment-5c4d4b7754-xc9qr 1/1     Running   0           9s
website-deployment-5c4d4b7754-xc9qb 1/1     Running   0           9s
ubuntu@ip-172-31-44-185:~/website$
```

i-Ob175896d7f855fcd (Worker-2)
PublicIPs: 13.60.52.77 PrivateIPs: 172.31.44.185

```
--> Using cache
--> 151e55a644a8
Successfully built 151e55a644a8
Successfully tagged deepan2003/website:v1
ubuntu@ip-172-31-44-185:~/website$ docker push deepan2003/website:v1
The push refers to repository [docker.io/deepan2003/website]
88d1d984b765: Mounted from library/nginx
5435b2dcdf5c: Mounted from library/nginx
054715a6dffa: Mounted from library/nginx
84e114c2bb36: Mounted from library/nginx
4a038fd18db1: Mounted from library/nginx
7b5d674621c2: Mounted from library/nginx
448ea5cac5d5: Mounted from library/nginx
98e94c91c91c: Pushed
v1: digest: sha256:151e55a644a8b6a880b3b500839a96db1c7f5b9c2a9fc586a5fb81f8a456e8 size: 2046
ubuntu@ip-172-31-44-185:~/website$ nano deployment.yaml
deployment.apps/website-deployment created
ubuntu@ip-172-31-44-185:~/website$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
website-deployment-5c4d4b7754-xc9qr 1/1     Running   0           9s
website-deployment-5c4d4b7754-xc9qb 1/1     Running   0           9s
ubuntu@ip-172-31-44-185:~/website$ nano service.yaml
service/website-service created
ubuntu@ip-172-31-44-185:~/website$ kubectl get svc
NAME      TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes ClusterIP  10.96.0.1     <none>        443/TCP          22m
website-service NodePort    10.105.47.92 <none>        80:30008/TCP     7s
ubuntu@ip-172-31-44-185:~/website$
```

i-Ob175896d7f855fcd (Worker-2)
PublicIPs: 13.60.52.77 PrivateIPs: 172.31.44.185

As per the requirement in the production server, you need to use the Kubernetes cluster and the containerized code from Docker Hub should be deployed with 2 replicas. Create a NodePort service and configure the same for port 30008

Hello world!



GitHub

Jenkins / devops-pipeline

- Status
- Changes
- Build Now
- Configure
- Delete Pipeline
- Full Stage View
- Stages
- Rename
- Pipeline Syntax

devops-pipeline

Add description

Stage View

Average stage times: (full run time: ~23s)		Clone Code	Build Docker Image	Push to DockerHub	Deploy to Kubernetes
		1s	11s	6s	2s
Apr 21 17:16	1 commit	1s	11s	6s	2s

Permalinks

- Last build (#3), 55 sec ago
- Last stable build (#3), 55 sec ago
- Last successful build (#3), 55 sec ago
- Last completed build (#3), 55 sec ago

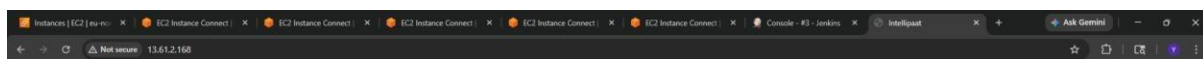
Builds >

Filter

Today: #3 11:46 AM


```
Instances | EC2 [eu-north-1] x EC2 Instance Connect [eu- x EC2 Instance Connect [eu- x EC2 Instance Connect [eu- x EC2 Instance Connect [eu- x EC2 Instance Connect [eu- x Console - #3 - Jenkins x + Ask Gemini x - x x
Not secure 13.61.2.168:8080/job/devops-pipeline/3/console
Jenkins / devops-pipeline / #3
054715a8bffa: Waiting
ca9dc5b98bb: Waiting
88d1d984b765: Waiting
543b2dcdf5c: Layer already exists
84e114c2bb36: Waiting
84e114c2bb36: Layer already exists
448ea5cac5d5: Layer already exists
054715a8bffa: Layer already exists
ca9dc5b98bb: Waiting
88d1d984b765: Layer already exists
ca9dc5b98bb: Pushed
v1: digest: sha256:d7f221958a199eab0245f6b4b2e760e9933b5e72bf2957478430849d6f8f909 size: 2046
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy to Kubernetes)
[Pipeline] sh
+ kubectl apply -f deployment.yaml
deployment.apps/website-deployment unchanged
[Pipeline] sh
+ kubectl apply -f service.yaml
service/website-service unchanged
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
Jenkins 2.555.1
```

SUCCESSFUL PIPELINE EXECUTION



WEBSITE OUTPUT